Reviewer Report

MCA

Manuscript number: mca-4499257

Manuscript title: Spectral Isomorphism between Renormalization Flow in Non-Autonomous Quadratic Maps and the Riemann Zeros: A Numerical Study

Review date: August 6, 2026

Recommendation: **Major Revision**

Overall assessment

The manuscript investigates a very interesting and ambitious problem: whether a non-autonomous nonlinear dynamical system can generate spectral features related to the non-trivial zeros of the Riemann zeta function. The authors construct a quadratic-map-based renormalization flow, introduce a logarithmic cooling mechanism, numerically extract spectral information from an empirical transfer operator, and compare the obtained eigenphase spectrum with the Riemann zeros.

The topic is highly interdisciplinary and potentially attractive to readers interested in dynamical systems, quantum chaos, random matrix theory, and the Hilbert–Pólya conjecture. The manuscript contains extensive numerical experiments and presents several interesting observations, including the low-order spectral alignment, the $N \approx 20$ residual feature, and the role of conjugate spectral symmetry.

However, in its current form, the manuscript has several fundamental issues regarding mathematical formulation, numerical methodology, parameter selection, and interpretation of results. Many conclusions appear to be based on numerical coincidence rather than demonstrated spectral correspondence. The authors repeatedly state that the results are heuristic, which is appropriate; nevertheless, several claims in the abstract, introduction, and conclusion remain stronger than what is supported by the presented evidence. Therefore, substantial revision is required.

Major comments

1. The mathematical definition of the spectral operator is unclear.

The manuscript states that the evolution is governed by a non-autonomous quadratic map

$$x_{n+1} = 1 - \lambda_n x_n^2,$$

with

$$\lambda_n = \mu_c + \frac{k_1}{\ln^2 n} + \frac{k_2}{\ln^3 n},$$

and then introduces a transfer matrix obtained from Gaussian kernel discretization.

However, the following points require clarification:

- What is the exact Hilbert space on which the operator acts?
- Is the transfer operator Markovian, unitary, or non-normal?
- How are eigenvalues related to physical phases?
- Why should the eigenphases of this non-unitary dissipative operator correspond to Riemann zeros?

The manuscript uses terminology from quantum mechanics (“unitary evolution”, “energy levels”, “spectral symmetry”), but the constructed operator appears to be a classical Perron–Frobenius-type transfer operator. The mathematical relationship between these two viewpoints must be established.

2. The connection to the Hilbert–Pólya conjecture is currently overstated.

The manuscript presents the method as a possible ”bottom-up” approach to the Hilbert–Pólya program. However, the Hilbert–Pólya conjecture requires a self-adjoint operator whose spectrum exactly reproduces the imaginary parts of the Riemann zeros.

The current model:

- is dissipative;
- depends on numerical discretization;
- requires Gaussian smoothing;
- requires parameter optimization;
- uses empirical linear scaling.

Therefore, it does not provide evidence for an underlying Hermitian operator.

The authors should clearly distinguish between:

1. reproducing statistical features of Riemann zeros;
2. fitting low-order zeros numerically;
3. constructing a genuine spectral realization.

At present these concepts are occasionally mixed.

Minor comments

1. The introduction is excessively long and should be shortened. The discussion of Hilbert–Pólya, random matrix theory, and quantum simulations could be made more focused.

2. The manuscript contains many speculative physical interpretations. These should be separated clearly from numerical observations.

3. The figures should include numerical uncertainty estimates.

4. The authors should report computational cost, including CPU/GPU information and memory consumption.

5. The reproducibility of the numerical experiments should be improved. Code availability would significantly strengthen the manuscript.

6. The conclusion should be rewritten to avoid statements suggesting a possible solution of the Hilbert–Pólya problem.

Final recommendation

This manuscript presents an interesting computational exploration connecting nonlinear dynamics, transfer operators, and the spectral properties of Riemann zeros. The idea of investigating non-autonomous chaotic systems as possible spectral generators is original and potentially valuable.

However, the current manuscript mainly demonstrates a numerical fitting phenomenon rather than a mathematically established spectral correspondence. The main challenges are the unclear operator-theoretic foundation, possible overfitting, insufficient convergence analysis, and overinterpretation of numerical coincidences.

$\boxed{\text{Major Revision}}$